LLMPhysics Journal Ambitions Contest: A Pre-Registered Study of Submission Quality

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Abstract—Large language models enable amateur physicists to produce work that closely resembles professional literature in form, while reducing the reliability of self-evaluation needed to improve its substance. This study presents a pre-registered quasi-experimental test of whether a lightweight structural incentive — a community contest with a publicly available rubric and recognition incentives — can produce a measurable improvement in submission quality in LLM-assisted amateur physics.

A baseline corpus of 50 papers from r/LLMPhysics was scored by two frontier models (GPT-5.2 and Claude Sonnet 4.6) under identical automated conditions, with scoring variance characterized via a 100-run noise floor. Contest submissions ($n = 10$) were evaluated using the same pipeline, and effect sizes were computed as noise-corrected Hedges' g' with 10,000-iteration bootstrap confidence intervals.

H_1 was supported across all six rubric categories and the normalized composite score. The composite yielded $g' = 1.33$ (95% CI: [0.60, 2.33]), with per-category effects ranging from 0.73 to 1.46. Baseline characterization additionally revealed systematic inter-model scoring divergence and non-zero variance at temperature 0.

These results indicate that structured community incentives can produce substantial improvements in the quality of LLM-assisted amateur scientific work. While the study does not directly measure the underlying reinforcement dynamics, the observed effects are consistent with interventions that counter known limitations in LLM-mediated self-evaluation.

1 Introduction

Amateur physicists with independently developed theories of the natural world are not a new phenomenon. Such theorists have sought out physicists for at least a century [Baez, 1998, Harris, 2012]. Writing from direct experience consulting with such individuals, Hossenfelder [2016] offers a portrait of a population driven by real intellectual investment rather than charlatanism: people who had, in many cases, independently reinvented territory already mapped by professional physics, without knowing the map existed.

The emergence of large language model chatbots has fundamentally changed what this population can produce. Where such theorists once worked with crude diagrams, informal notes, and occasional elementary mathematics [Hossenfelder, 2016], LLMs now enable the rapid development of theories in full formal register: complete with Lagrangians, field metrics, structured abstracts, and LaTeX formatting [Siegel, 2025, Collier, 2025]. The result is output that closely resembles professional physics literature in appearance — a change that presents new challenges not only for outside observers attempting to distinguish it from genuine work, but for the authors themselves. The effects of this have begun to reach institutional levels: arXiv, the preprint server that has hosted physics and mathematics re-

search since 1991, announced in March 2026 that it would separate from Cornell University and restructure as an independent nonprofit [Brainard, 2026]. In that report, arXiv leadership framed the shift as a way to broaden funding support and finance staffing and technology during rapid submission growth, alongside mounting concerns about AI-generated submissions. Online communities have also begun springing up, such as r/LLMPhysics — founded in May 2025 and, as of March 24, 2026, home to roughly 5,000 members dedicated entirely to receiving and discussing this output [r/LLMPhysics, 2025, 2026]. As the production process is increasingly delegated to a model whose outputs are fluent and authoritative in tone, the author's own ability to evaluate what has been produced becomes progressively harder to calibrate [Cheng et al., 2025, Siegel, 2025, Collier, 2025].

This erosion of self-evaluation is not incidental — it has a documented mechanism. Sycophancy is a systematic feature of RLHF-trained language models: analyzing 15,000 human preference comparisons, Sharma et al. [2024] found that the resulting preference model selected sycophantic responses over truthful corrections 95% of the time.

Batista and Griffiths [2026] demonstrate that LLM interaction can increase a user's confidence in a hypothesis without increasing its truth value. In their experiment, default LLM behavior was shown to be statistically equivalent to explicitly sycophantic prompting on confidence inflation — and participants receiving unbiased feedback discovered the true rule far more often, becoming less confident over time rather than more so. The implication is not merely that sycophantic AI inflates confidence, but that default AI interaction systematically substitutes for the disconfirming evidence that would naturally correct it. Users prefer sycophantic responses, rate them as higher quality, and are more likely to return to the systems that produce them — even as those interactions increase conviction, reduce willingness to repair mistaken positions, and erode accurate self-assessment [Rathje et al., 2025, Cheng et al., 2025]. This tendency is not confined to conversational exchanges. When prompted to prove mathematical statements that are false but plausible, frontier models will typically construct apparently rigorous proofs rather than identify the error — defaulting, even in formal reasoning, toward agreement with the user's implicit premise [Dekoninck et al., 2026].

On March 1, 2026, the moderator u/AllHailSeizure announced the LLMPhysics Journal Ambitions Contest on r/LLMPhysics [AllHailSeizure, 2026], a community contest designed to encourage higher-quality submissions through structured recognition incentives. The contest was conceived and announced by u/AllHailSeizure as a community initiative; the present study's contribution is the pre-registered measurement framework built around it: a quasi-experimental design using automated LLM scoring against an empirically characterized noise floor to test whether a lightweight structural incentive can produce a measurable shift in output quality within precisely this population — one subject to the sycophantic reinforcement dynamics described above.

2 Methods

This study employs a pre-registered statistical framework to evaluate whether structured contest incentives produce a measurable improvement in submission quality relative to a pre-established baseline. Throughout this paper, "quality" refers strictly to performance on the six rubric dimensions described in Section 2.2; no claims are made about quality outside those criteria. All analysis thresholds, equations, and decision rules were fixed prior to the scoring of any paper.

A note on the interpretation of scores is warranted. The quality measures reported in this study represent the output of the scoring models under the specified prompt and configuration — they are not claims about the true underlying quality of any paper. A high score in Scientific Humility, for instance, reflects that the scoring model produced that output given the rubric criteria; it does not assert that the paper genuinely demonstrates intellectual restraint. The scores are treated throughout as measurements of a defined operationalization, not as ground truth evaluations. This distinction applies equally to both models and to all six categories.

2.1 Corpus Collection

The baseline corpus consists of 50 papers drawn from r/LLMPhysics [r/LLMPhysics, 2025], a Reddit community in which amateur researchers post physics theories developed with LLM assistance. Papers were collected by scrolling backwards through the subreddit's post history, covering the window January 9, 2026 to February 26, 2026. This window represented a larger available pool; papers were selected opportunistically rather than exhaustively. Posts were included if they made a recognizable attempt to present a physics theory, however underdeveloped. Posts were excluded if they were satirical, meta-commentary, single-line, below a minimum content threshold, or hosted behind access-restricted sites.

The corpus comprised mixed file formats: 21 PDFs, 3 Markdown files, and 26 plain-text transcriptions manually copied from Reddit posts. All files were stored locally and processed uniformly by the scoring pipeline.

The contest sample consisted of all valid submissions received during the contest window, with size n_1 determined at close of submissions.

2.2 Scoring Rubric

The rubric was designed primarily by u/AllHailSeizure and reviewed collaboratively by the broader team. It evaluates six categories with the following maximum point allocations: Hypothesis (15), Novelty (15), Scientific Humility (15), Engagement (20), Rigor (10), and Citations (10), for a raw maximum of 85 points. The normalized total score was computed as:

$$S_{\text{norm}} = \frac{S_{\text{raw}}}{85} \times 100 \tag{1}$$

Statistical analysis was conducted on each of the six category scores individually and on S_{norm}, yielding seven parallel analyses.

It should be noted that the human judging panel additionally scored a Defense category (15 points) not available to the automated pipeline, as it requires live engagement with post-submission critique. The normalized formula accounts for this structural difference between human and automated evaluation.

2.3 Scoring Prompt

Each paper was scored using a standardized judge prompt, guidelines_v1.1, which instructed the model to assume the role of a contest judge with no function outside the rubric criteria. The prompt provided per-category scoring anchors — full marks, partial credit, and low marks — with specific criteria for each, and enforced strict JSON-only output containing a numeric score and one-to-two sentence justification per category. No explanatory text, markdown formatting, or commentary outside the JSON structure was permitted.

The prompt was stored as a plain-text file on disk and loaded fresh at the start of every individual scoring call, rather than being embedded in the pipeline code. Its integrity was verified by a SHA-256 hash recorded in the

project repository prior to any scoring run, establishing that the scoring instructions were identical across all runs and were not modified after scoring began.

2.4 Automated Scoring Pipeline

Scoring was conducted via API calls to two large language models: GPT-5.2 (OpenAI) and Claude Sonnet 4.6 (Anthropic). An API (Application Programming Interface) call is a programmatic request sent directly to a model provider's servers, bypassing any chat interface. Both models were configured identically: temperature 0, meaning the model deterministically selected the highest-probability token at each step rather than sampling from a weighted distribution [Atil et al., 2025], and a fresh context window for every individual scoring run with no conversation history carried between papers or models.

Prior to each API call, papers were prepared according to file type. PDFs were base64-encoded — a method of representing binary file data as plain text — and transmitted as document objects, preserving the file's original structure including mathematical notation. Plain-text and Markdown files were transmitted as text. This approach was chosen over text extraction to avoid mangling equations and formatting during conversion.

Each paper was scored independently by both models in sequence. After each paper, the pipeline paused for 60 seconds to respect API rate limits. Each scoring call included up to three attempts with a 5-second pause between retries. If all three attempts failed, the run was logged as an error record rather than silently discarded, with the failure reason captured for review.

Every successful scoring run produced a structured JSON record containing the paper identifier, citation, group label (baseline or contest), model, UTC timestamp, token counts, and the model's scores and justifications for all six categories.

Prior to group-level statistical analysis, per-paper scores were averaged across both models. Papers for which either model produced an error record were excluded from averaged statistics. Bootstrap resampling operated on these averaged per-paper values.

2.5 Pipeline Limitations

Two baseline papers initially produced error records due to API rate limits on the Anthropic API tier used for this study. A targeted rerun script was written to scan all output files for error records, re-attempt failed runs with extended pauses between calls, and overwrite failed records in place upon success. Both papers were successfully scored on rerun. As the failures were attributable solely to API rate limiting rather than any methodological issue, they are not treated as exceptions to the pre-registered procedure.

2.6 Noise-Corrected Effect Size Equations

For reproducibility, we state the exact pre-registered equations used to compute the noise-corrected effect size in each category c (and for S_{norm}). Observed pooled variance and standard deviation were computed as:

$$s^2_{\text{pooled},c} = \frac{(n_1 - 1)s^2_{1,c} + (n_2 - 1)s^2_{2,c}}{n_1 + n_2 - 2} \quad (2)$$

$$SD_{\text{observed},c} = \sqrt{s^2_{\text{pooled},c}} \quad (3)$$

Noise-corrected variance and standard deviation were then:

$$s^2_{\text{corrected},c} = SD^2_{\text{observed},c} - \sigma^2_{\text{noise},c} \quad (4)$$

$$SD_{\text{corrected},c} = \sqrt{s^2_{\text{corrected},c}} \quad (5)$$

For figure error bars, we plot $\pm\sigma_{\text{noise},c}$ where $\sigma_{\text{noise},c} = \sqrt{\sigma^2_{\text{noise},c}}$, using the noise-floor estimates.

The raw standardized difference and Hedges-corrected effect [Hedges, 1981] were:

$$g_c = \frac{\bar{X}_{1,c} - \bar{X}_{2,c}}{SD_{\text{corrected},c}} \tag{6}$$

Throughout, subscript 1 denotes the contest group and subscript 2 denotes the baseline group, such that positive g'_c values indicate higher contest scores. Here, $\bar{X}_{1,c}$ and $\bar{X}_{2,c}$ are group means computed from the model-averaged per-paper values in category c.

$$J = 1 - \frac{3}{4(n_1 + n_2 - 2) - 1} \tag{7}$$

$$g'_c = J\, g_c \tag{8}$$

To quantify the sensitivity of effect sizes to noise subtraction, we also computed an uncorrected comparator using the observed pooled standard deviation:

$$g'_{\text{uncorrected},c} = J\, \frac{\bar{X}_{1,c} - \bar{X}_{2,c}}{SD_{\text{observed},c}} \tag{9}$$

The corrected and uncorrected estimates were then compared category-wise and for S_{norm} (reported in Section 3.6).

Edge-case handling followed the pre-registration exactly: if $s^2_{\text{corrected},c} \leq 0$, the effect estimate for category c was treated as invalid (no real-valued $SD_{\text{corrected},c}$). Bootstrap confidence intervals used paper-level resampling with replacement within each group (contest: n_1 papers; baseline: $n_2 = 50$ papers) on the model-averaged per-paper values, recomputing g'_c per replicate while holding $\sigma^2_{\text{noise},c}$ fixed. If more than 10% of replicates were invalid for a category, that category's CI was declared unreliable.

2.7 Hypotheses and Decision Framework

Consistent with `methodology_v1_1.pdf`, decision testing was conducted per category c (and for S_{norm}) using a three-region pre-registered framework with a practical null zone:

$$H_0 : |g'_c| < 0.2 \tag{10}$$

$$H_1 : g'_c \geq 0.2 \tag{11}$$

$$\text{Adverse-effect region: } g'_c \leq -0.2 \tag{12}$$

Outcome classification used three exhaustive, mutually exclusive outcomes together with validity checks from Section 2.6:

- *Practical-null/invalid outcome* (reported as fail to reject H_0) when $\sigma^2_{\text{noise},c} \geq SD^2_{\text{observed},c}$ (invalid corrected denominator) or when $|g'_c| < 0.2$.
- *Favorable-effect outcome* (support H_1) when $g'_c \geq 0.2$.
- *Adverse-effect outcome* when $g'_c \leq -0.2$.

2.8 Data and Code Availability

All study materials are available in the public project repository: `https://github.com/AllHailSeizure/LLMPhysics-Journal-Ambitions-Contest`. The repository is complete and includes the manuscript source, generated figures, bibliography, pre-registration audit materials, scoring pipeline code, baseline and contest corpora, raw JSON scoring outputs, SHA-256 audit artifacts, and extended per-paper scoring-justification appendices (Appendix A for the full scoring prompt, Appendix B for baseline papers, and Appendix C for contest papers).

3 Results

This section reports both baseline corpus characterization and contest arm outcomes under the pre-registered framework, including noise-corrected Hedges' g', bootstrap confidence intervals, and outcome classifications.

3.1 Baseline Score Distributions

Across both models, baseline scores were low relative to the rubric maxima, and the S_{norm} distribution was right-skewed in both cases. Claude Sonnet 4.6 produced a mean S_{norm} of 24.3 ($\sigma = 15.8$, median = 20.9), while GPT-5.2 produced a mean of 32.0 ($\sigma = 15.6$, median = 29.4). Despite the offset in absolute scores, the spread of quality was nearly identical between the two models, with standard deviations differing by less than 0.2 points. The right skew — visible in both distributions in Figure 1 — indicates that most papers clustered in the lower score ranges, with a smaller number of higher-scoring papers pulling the mean above the median in both cases.

The breadth of the distribution was itself a notable finding. Given the opportunistic nature of the sampling procedure, it was not clear in advance whether the corpus would capture genuine quality variation or simply reflect a narrow band of uniformly poor submissions. The observed range — Claude: 2.4 to 73.5; GPT: 7.1 to 73.5 — and the roughly normal shape of the bulk of both distributions suggest the corpus captured meaningful variance across the subreddit's output, lending confidence to its use as a baseline reference population.

3.2 Per-Category Baseline Performance

Averaged across both models, all six rubric categories scored below 50% of their respective maxima, and three categories — Citations, Rigor, and Engagement — showed particularly weak performance (Figure 2). Citations averaged 1.75 out of 10 across both models, with Claude's median of 0.25 and a modal score of 0.0 indicating that the majority of baseline papers either lacked citations entirely or provided only token references. Rigor averaged 2.88 out of 10, and Engagement averaged 3.58 out of 20.

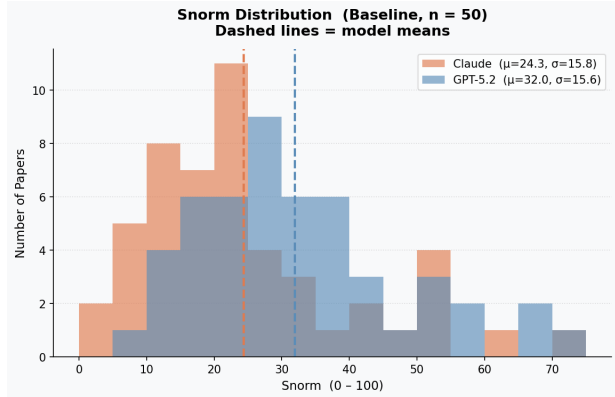

Figure 1: S_{norm} distributions for the baseline corpus ($n = 50$) by model. Dashed lines indicate model means. Both distributions are right-skewed, with the majority of papers scoring below 40.

The Citation finding warrants specific comment. Many baseline papers included references that appeared as isolated lists, if any references were included at all, rather than sources integrated into specific claims. Under the rubric criteria, which reward citations tied directly to supporting arguments, this pattern yields low scores.

Scientific Humility was the relative strength of the baseline corpus, averaging 6.45 out of 15 across both models, though this still represents less than half the available points. The radar plot in Figure 3 illustrates the overall profile clearly: both models place the baseline corpus well within the inner 40% of the rubric space across all dimensions, with no category approaching its maximum.

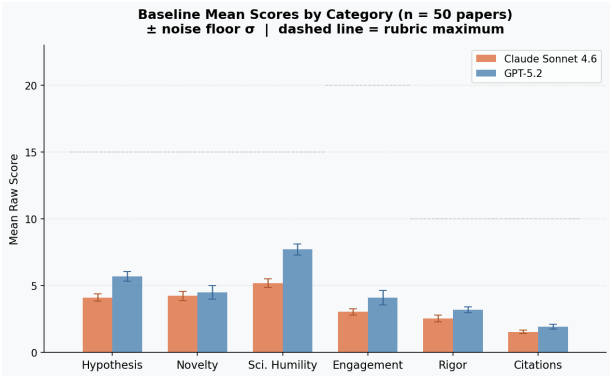

Figure 2: Mean raw scores per category by model for the baseline corpus ($n = 50$), with $\pm\sigma_{\text{noise},c}$ error bars ($\sigma_{\text{noise},c} = \sqrt{\sigma^2_{\text{noise},c}}$). Dashed lines indicate rubric maxima.

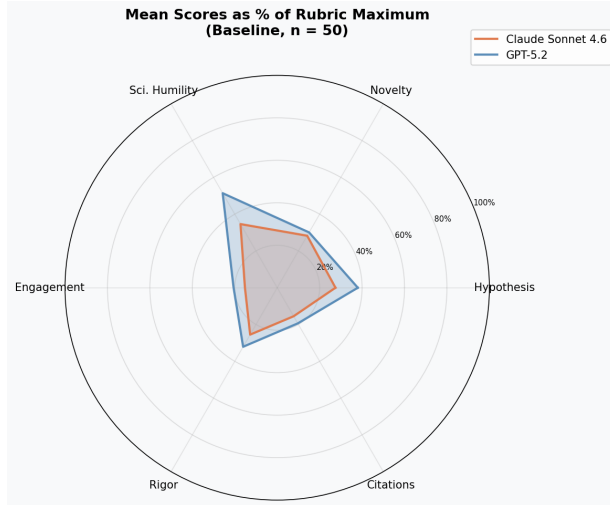

Figure 3: Mean scores as percentage of rubric maximum per category, baseline corpus ($n = 50$). Both models place the corpus predominantly within the inner 40% of the rubric space. A notable exception is GPT-5.2 on Scientific Humility, which crosses the 40% threshold — the same category showing the largest inter-model divergence, with Claude scoring a mean of 2.52 points lower than GPT on that dimension.

3.3 Inter-Model Divergence

Claude Sonnet 4.6 scored lower than GPT-5.2 on every rubric category across all 50 baseline papers, a finding that was systematic rather than incidental. Of the 50 papers, GPT assigned a higher S_{norm} on 44, Claude on 3, with 3 papers producing equal scores (Figure 4). The mean S_{norm} delta was -7.6 points (Claude minus GPT), with a standard deviation of 6.2. Despite this offset, the paper-level scatter plot shows points clustering tightly around the OLS regression line ($y = 0.93x - 5.48$), indicating that both models largely agreed on the relative quality ordering of papers even while disagreeing on absolute scores. The two models appear to be measuring the same underlying construct, but with a consistent calibration difference.

The largest per-category divergence occurred in Scientific Humility, where Claude scored a mean of 2.52 points lower than GPT — representing approximately 17% of the category maximum (Figure 5). Hypothesis showed the second largest gap at -1.58 points, while Novelty showed the smallest divergence at -0.27 points.

The source of this systematic divergence cannot be determined from the available data. One plausible confound is that GPT-5.2 exposes a reasoning effort parameter not available in the Anthropic API, meaning the two models were not perfectly matched in inference configuration despite identical temperature and context settings. Whether adjusting this parameter would reduce the inter-model gap remains a question for future work. More broadly, both models are proprietary systems whose internal architectures are not fully disclosed, placing a hard limit on the interpretive conclusions that can be drawn about the cause of the divergence.

Importantly, the systematic nature of the divergence does not undermine the contest arm analysis, but it does bound interpretation. Both the baseline and contest corpora were scored by the same models under identical conditions, so stable model-

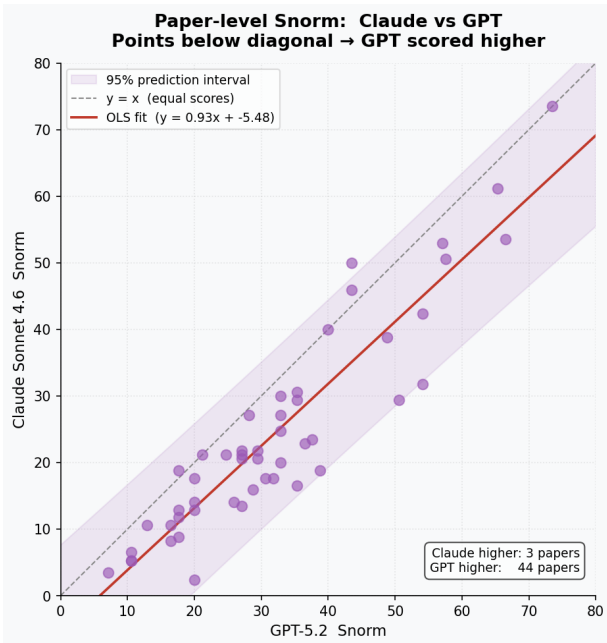

Figure 4: Paper-level S_{norm} scores: Claude Sonnet 4.6 vs GPT-5.2. Points below the diagonal indicate GPT scored higher. The OLS fit ($y = 0.93x - 5.48$) and 95% prediction interval are shown. GPT scored higher on 44 of 50 papers.

level calibration offsets should be partially controlled in the between-group comparison. However, category-specific divergence can still vary across groups — as seen in Scientific Humility — so this should not be interpreted as complete cancellation of calibration bias.

3.4 Noise Floor Results

The noise floor characterization confirmed that temperature 0 does not fully eliminate scoring variance. Across all six categories, $\sigma^2_{\text{noise},c}$ was empirically non-zero for both models, with GPT-5.2 exhibiting substantially higher within-paper variance than Claude Sonnet 4.6 on most dimensions (Figure 6). The largest pooled noise variances were observed in Novelty ($\sigma^2_{\text{noise}} = 0.194$) and Engagement ($\sigma^2_{\text{noise}} = 0.171$), both categories requiring interpretive judgment about the paper's relationship to existing literature and the field. Citations showed the lowest noise variance ($\sigma^2_{\text{noise}} = 0.027$), consistent with its more quantitative and verifiable nature.

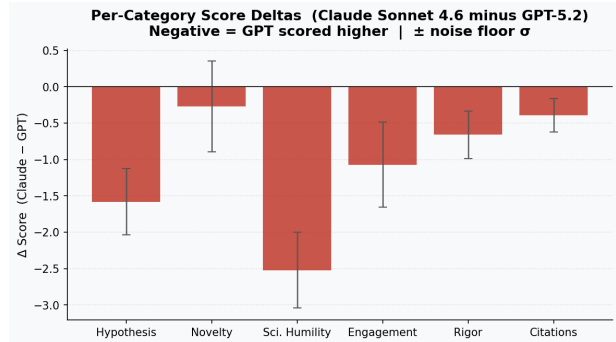

Figure 5: Per-category mean score deltas (Claude minus GPT) with $\pm\sigma_{\text{noise},c}$ error bars. All categories are negative, indicating Claude scored lower on every dimension. Scientific Humility shows the largest divergence.

GPT-5.2's higher variance was most pronounced in Novelty ($\sigma^2 = 0.270$ vs Claude's 0.118) and Engagement ($\sigma^2 = 0.288$ vs Claude's 0.055). A plausible explanation involves the reasoning effort parameter available in the GPT-5.2 API, which may introduce additional stochasticity in how the model approaches interpretive judgments across repeated runs. However, as with the inter-model score divergence, this remains speculative given the proprietary nature of both systems.

The noise floor values are consequential for the contest arm analysis. Under the preregistered estimator in Section 2.6, subtracting $\sigma^2_{\text{noise},c}$ from pooled variance can reduce the denominator and therefore slightly increase $|g'_c|$ when the corrected variance remains positive. To quantify this potential inflation directly, we ran a corrected-versus-uncorrected sensitivity analysis (Section 3.6; Figure 9). In this dataset, the correction shifted effect sizes only minimally and did not change any outcome classification.

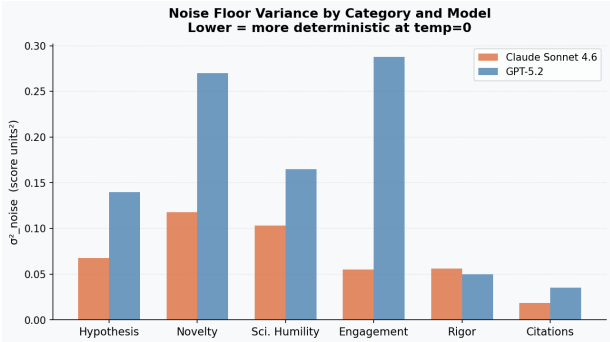

Figure 6: Noise floor variance (σ^2_{noise}) by category and model. GPT-5.2 exhibits higher variance than Claude Sonnet 4.6 on most dimensions, particularly Novelty and Engagement. Citations show the lowest variance across both models.

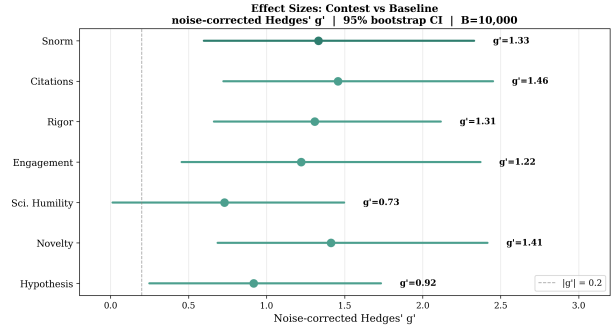

Figure 7: Noise-corrected Hedges' g' by category and composite score for contest vs. baseline, with 95% bootstrap confidence intervals. All seven analyses clear the pre-registered detectability criterion on the basis of point estimates; Scientific Humility's CI straddles 0.2 and is interpreted with additional caution (see text).

3.5 Implications for the Contest Arm

Taken together, the baseline results establish three things relevant to the contest arm analysis. First, the baseline corpus represents a low-quality reference population by the rubric's standards, meaning there is substantial headroom for improvement — a ceiling effect is unlikely to suppress detectable contest effects. Second, the per-category profile suggests that Citations, Rigor, and Engagement are the dimensions where contest incentives are most likely to produce measurable improvement, as these categories have clear and actionable criteria that participants can directly address. Scientific Humility and Novelty, being more qualitative and harder to target deliberately, may show weaker or more inconsistent effects. Third, the non-trivial noise floor — particularly in Novelty and Engagement — makes explicit sensitivity checks necessary because corrected effect sizes can shift upward when noise is subtracted. In this study, that shift was empirically negligible and did not materially alter inference.

3.6 Contest Arm Results

The contest arm analysis was conducted on $n_1 = 10$ submissions received during the contest window. Effect sizes, confidence intervals, and outcome classifications were computed per the pre-registered decision framework described in Section 2, with $\sigma^2_{\text{noise},c}$ held fixed at the values established during baseline characterization. All contest-versus-baseline group statistics in this subsection are computed from model-averaged per-paper scores, as specified in Section 2.6.

The overall result was consistent across all seven pre-registered analyses. H_1 was supported for the six individual rubric categories and S_{norm}, with all noise-corrected Hedges' g' point estimates meeting the pre-registered improvement threshold ($g' \geq 0.2$). The 95% bootstrap confidence intervals ($B = 10{,}000$) were above zero in all seven analyses; six of seven were entirely above 0.2, while Scientific Humility straddled the 0.2 boundary and is therefore interpreted with additional caution. The mean S_{norm} rose from approximately 28.2 in the baseline corpus to approximately 48.9

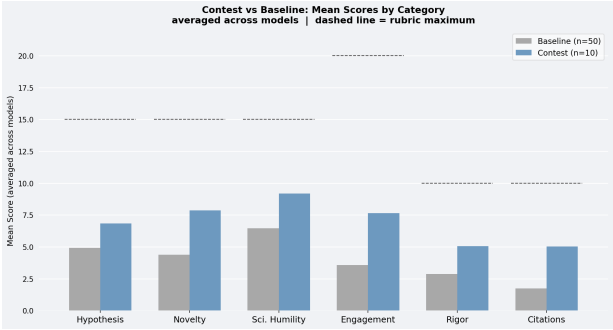

Figure 8: Mean raw category scores for baseline and contest groups, by model, with $\pm\sigma_{\text{noise},c}$ error bars. Contest means are higher than baseline means across all rubric dimensions.

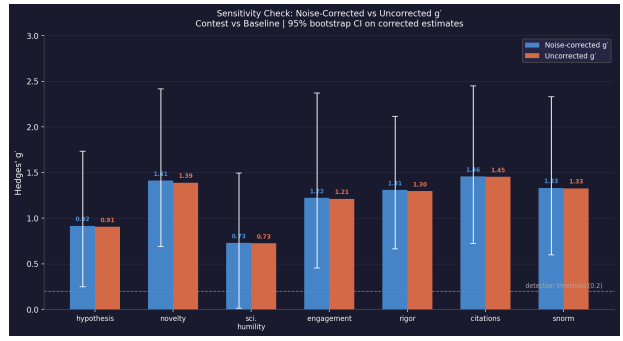

Figure 9: Sensitivity analysis comparing uncorrected and noise-corrected Hedges' g'. Subtracting the noise floor produces only minimal upward shifts, with no change in outcome classification for any category or for S_{norm}.

in the contest arm, a raw difference of approximately 20.7 normalized points. The standard deviations of the two groups were nearly identical (baseline: $\sigma = 15.4$; contest: $\sigma = 15.7$), indicating that the shift reflected a broad upward movement of the distribution rather than a small number of outlying high-scorers. The S_{norm} composite yielded $g' = 1.33$ (95% CI: [0.60, 2.33]), classified as a large improvement under the pre-registered magnitude conventions.

Because subtracting $\sigma^2_{\text{noise},c}$ can reduce the denominator and inflate effect sizes, we ran the explicit sensitivity check defined in Section 2 by comparing corrected g' to uncorrected $g'_{\text{uncorrected}}$ (Figure 9). The impact was negligible: for S_{norm}, g' changed from 1.328 (uncorrected) to 1.332 (corrected), $\Delta g' = +0.004$. Across the six rubric categories, absolute shifts ranged from 0.004 to 0.023, and all seven outcome classifications were unchanged.

The magnitude of this result was larger than anticipated. Given the overall structure of the methodology — especially bootstrap confidence intervals with $n_1 = 10$ and known inter-model calibration differences — the prior expectation was a small or at most medium effect in the most actionable categories, with Novelty and Engagement potentially failing to clear the detectability threshold. Every category cleared it.

Per-category effects. Figure 7 presents the full forest plot of per-category g' values with 95% bootstrap confidence intervals. Effect sizes ranged from $g' = 0.73$ in Scientific Humility (95% CI: [0.01, 1.50]) to $g' = 1.46$ in Citations (95% CI: [0.72, 2.45]). The pre-registered prediction that Citations, Rigor, and Engagement would show the strongest response — on the grounds that their criteria are concrete and directly actionable — was partially borne out: Citations ranked first and Rigor third. Novelty, however, ranked second at $g' = 1.41$ (95% CI: [0.69, 2.42]), despite having been identified as a qualitative category less amenable to deliberate targeting. One plausible account is that the six rubric categories are not independent constructs: authors who engaged seriously with the literature and scoped their empirical claims accordingly may have produced hypotheses that read as more genuinely novel as a downstream consequence. The available data do not permit this account to be distinguished from alternatives, and it is offered as a direction for future work rather than a confirmed mechanism. Figure 8 shows mean raw scores per category for both groups, and Figure 10 illustrates the shift in rubric

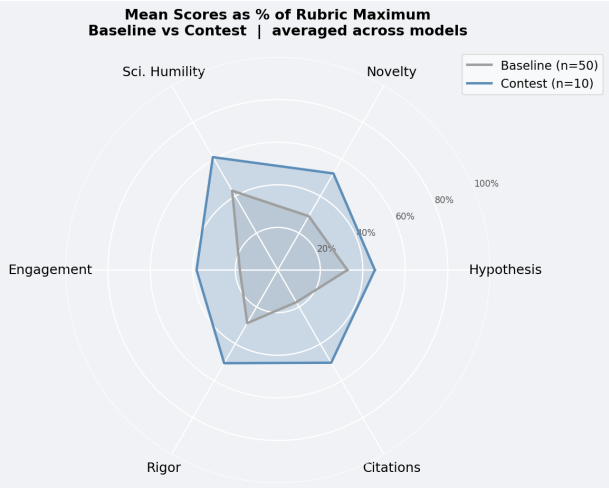

Figure 10: Mean scores as a percentage of category maxima for baseline vs. contest groups, by model. The contest profile shifts outward across the full rubric space.

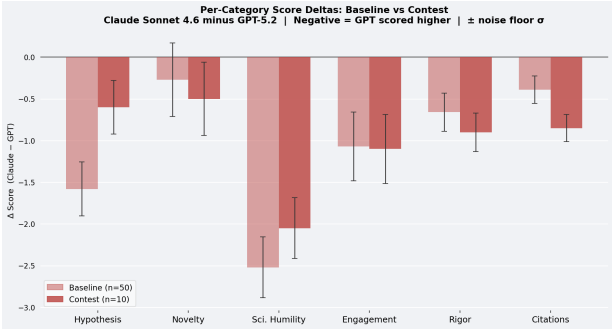

Figure 11: Claude-minus-GPT per-category mean deltas in baseline and contest groups. The largest widening of inter-model divergence appears in Scientific Humility.

profile from baseline to contest.

Inter-model divergence in the contest arm. The systematic divergence between Claude Sonnet 4.6 and GPT-5.2 documented in Section 3.3 persisted in the contest arm and widened in at least one category. As shown in Figure 11, Scientific Humility showed a larger Claude-minus-GPT gap in the contest arm than in the baseline, suggesting that the two models responded differently to contest submissions on that dimension specifically. A content-level account is consistent with this pattern: contest participants, motivated by recognition incentives, may have produced more performative epistemic restraint — hedging language and humility markers adopted instrumentally rather than reflectively — and Claude may be more sensitive than GPT to that distinction. This interpretation is supported by qualitative review of the scoring justifications in the repository appendices (Appendix B for baseline papers and Appendix C for contest papers; `https://github.com/AllHailSeizure/LLMPhysics-Journal-Ambitions-Contest`) but cannot be confirmed from the quantitative data alone. Importantly, this widening divergence does not undermine the main result: H_1 was supported on Scientific Humility despite Claude applying what appears to be a stricter standard, meaning the effect cleared a higher bar than GPT alone would have set.

Contest papers and inter-model agreement. A secondary observation visible in Figure 12 is that contest papers cluster more tightly around the equal-scores diagonal than baseline papers. Of the 50 baseline papers, GPT assigned a higher S_{norm} on 44; of the 10 contest papers, GPT scored higher on 9, with one contest paper scoring higher under Claude. The narrowing of the inter-model gap in the contest arm is consistent with the rubric-targeting account: when authors write deliberately toward specific criteria, both models have less interpretive ambiguity to resolve, reducing the surface area for calibration differences to express themselves.

Distribution shape and self-selection. The contest S_{norm} distribution (Figure 13) occupied the range approximately 18–70, with no papers clustering at the very bottom of the scale. Two contest submissions scored poorly in absolute terms, and the highest-scoring paper in the entire combined corpus was a baseline paper rather than a contest

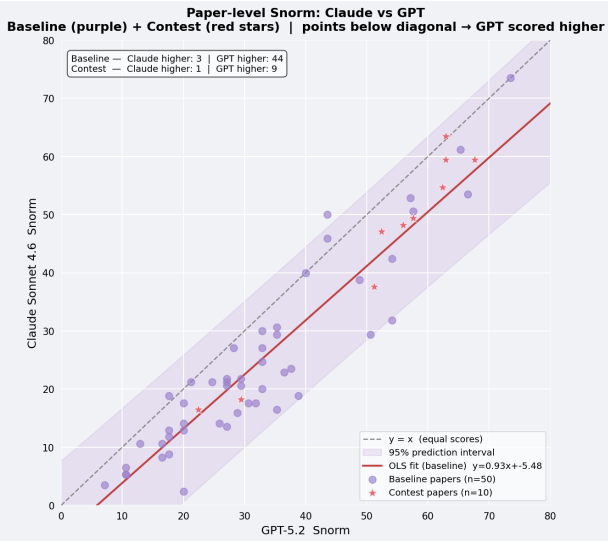

Figure 12: Paper-level S_{norm} scatter (Claude vs. GPT) with contest papers overlaid on baseline. Contest points cluster more tightly near the equal-scores diagonal, indicating reduced inter-model disagreement.

submission, underscoring that the contest did not produce a categorically separate quality tier. The two groups overlap substantially, and the contest effect is better described as a distributional shift than as a clean separation. It bears noting that contest participants are not a random draw from the subreddit population: self-selection into a formal evaluation context may correlate with effort, confidence, or familiarity with academic conventions, and this constitutes a genuine threat to the external validity of the effect size estimates. This limitation is addressed further in Section 4.2.

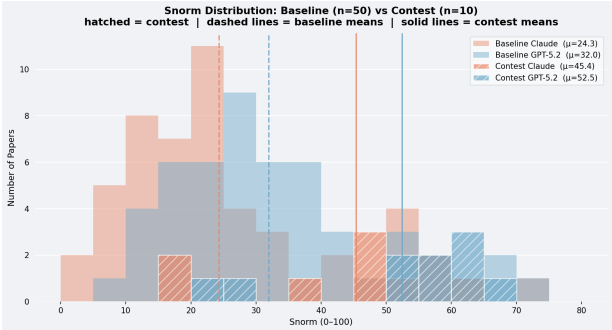

Figure 13: S_{norm} distributions for baseline and contest groups. The contest distribution is right-shifted while still overlapping the baseline range.

4 Discussion

4.1 Interpretive Summary

The central finding of this study is straightforward: a lightweight structural incentive — a community contest with a publicly available rubric and recognition incentives — produced a large, consistent improvement in submission quality relative to the baseline corpus, across all six rubric dimensions and the normalized composite score. The mean S_{norm} nearly doubled, and H_1 was supported in every pre-registered analysis. For this instance, the contest succeeded in its stated goal.

The result is notable in context. The baseline corpus established that the typical r/LLMPhysics submission scored below 50% of each rubric-category maximum, and the sycophancy literature reviewed in the Introduction provides a mechanistic account of why: LLM-assisted authors are being actively reinforced toward confidence and away from the disconfirming feedback that would naturally improve their work. Batista and Griffiths [2026] report that default model behavior produced substantially greater confidence increases than rule-disconfirming feedback ($d = 0.87$, 95% CI [0.58, 1.15]). The present study does not measure whether that

confidence dynamic was reduced — contest participants may remain just as confident in their output regardless of score outcomes. What the data do show is that a lightweight structural incentive produced a large improvement in submission quality ($g' = 1.33$) within a population operating under precisely those conditions. That the incentive package was sufficient to shift output quality — with a pre-registered, sensitivity-checked methodology, a small contest sample, and no control over how participants used their models — is the substantive finding.

The observed improvement is consistent with an effect of the contest package, under the confounding constraints acknowledged in Section 4.2. What cannot be determined is which component of the contest package was load-bearing. The rubric, the contest framing, the recognition incentive, and the self-selection of willing participants are fully confounded. A plausible account is that the public availability of the rubric allowed participants to incorporate its criteria directly into their model environment, and that the desire to perform well in a formal evaluation context increased the effort invested. The two contest submissions that scored poorly are a useful reminder that neither of these mechanisms is universal.

The broader significance of this finding extends beyond the subreddit context. The sycophancy literature cited in the Introduction [Sharma et al., 2024, Batista and Griffiths, 2026, Rathje et al., 2025, Cheng et al., 2025] is not studying populations with specific vulnerabilities. It is studying ordinary people using widely available tools, facing the default conditions of LLM interaction: confidence inflation, erroneous validation of invalid reasoning, and preference for agreement over correction. These conditions apply broadly, and the communities they produce — of which r/LLMPhysics is one example — are not going away. Access to powerful language models is not receding, and the curiosity that drives people to use them to pursue the "big questions" is a feature of human intellectual life, not a pathology to be eliminated.

The institutional response to date has been largely reactive. Brainard [2026] describes arXiv's March 2026 governance restructuring as a funding and infrastructure response to rapid preprint growth, with leadership also citing concern about AI-generated submissions as part of the operational pressure. The present study suggests a complementary direction: proactive community-level incentives that channel intellectual excitement toward better epistemic habits rather than attempting to quarantine or dismiss it. The contest did not produce professional physics. It produced meaningfully better amateur physics — work that engaged more seriously with the literature, scoped its claims more carefully, and demonstrated greater awareness of its own limitations. Whether this approach is scalable, durable, or generalizable to other communities remains an open question that warrants further investigation. What the present study establishes is that the attempt had an effect, and that the effect was large.

4.2 Limitations and Threats to Validity

Several limitations constrain the conclusions available from this study.

The most significant methodological limitation is the opportunistic nature of the baseline sampling procedure. Papers were selected by scrolling backwards through the subreddit's post history without a pre-specified sampling frame, which means the baseline corpus cannot be guaranteed to be representative of the subreddit's typical output. The effect size estimates are valid relative to this baseline, but their generalizability depends on an assumption of representativeness that cannot be formally verified.

Self-selection is a structural feature of the design rather than an oversight: a contest is self-selected by definition. Participants willing to submit work for formal public evaluation may differ systematically from the broader subreddit population in effort, confidence, or familiarity with academic conventions. This limits the external validity of the effect size estimates and should be interpreted accordingly.

The intervention components cannot be decomposed. The rubric, the contest structure, and the recognition incentive were presented as a package, and the available data provide no basis for attributing the observed effect to any one component. This is an inherent constraint of naturalistic quasi-experimental designs: the ecological validity that makes the study interesting is precisely what makes clean decomposition unavailable. This is acknowledged as a boundary on causal inference rather than a correctable flaw.

This study was designed around an existing community event rather than conceived independently. The contest was announced by u/AllHailSeizure prior to the present authors' involvement; the measurement framework was pre-registered and co-announced with the contest, and the baseline corpus was collected without any contest framing visible to the community. The pre-registration audit trail — SHA-256 hashes of both the methodology document and the scoring prompt, committed to the repository before any scoring began — provides independent verification that the analytical framework was not adjusted in response to results. Nevertheless, the intervention design itself was not under experimental control, which constrains the causal language available.

Finally, both scoring models are proprietary systems whose internal architectures are not fully disclosed. This places a hard limit on interpretive conclusions about the source of inter-model divergence and applies equally to every study of this kind in the current literature.

4.3 Future Work

Some limitations of this study are baked into the naturalistic design and would not disappear with a better-resourced replication — they are the cost of studying a real community event rather than a laboratory analog. What a follow-up study could improve is the degree to which the measurement framework and the intervention are designed together from the outset, rather than the former being built around the latter after the fact.

First, a follow-up study designed with this goal in mind would benefit from a pre-specified sampling frame for baseline collection, established before any papers are gathered. It would also benefit from a participant questionnaire administered at submission — capturing self-assessed confidence before and after drafting, familiarity with the rubric criteria, and whether the rubric was incorporated directly into the model environment. Such data would not fully decompose the intervention, but would provide crude proxies for rubric exposure versus contest-framing effects, partially addressing the confounding problem without sacrificing the naturalistic setting.

Second, a parallel methodological priority is improved characterization of scoring noise. The initial variance-subtraction approach used here was adequate for a first-pass correction and did not alter qualitative conclusions in sensitivity checks, but a richer noise model could better calibrate effect-size magnitudes and uncertainty estimates.

Third, an open question is durability. The present study captures a single contest window. Whether repeated interventions — semi-annual contests, for instance — would produce a cumulative shift in the subreddit's baseline submission quality is a question that would require retrospective corpus analysis across

multiple contest cycles to answer.

Conclusion

This study tested whether a lightweight structural incentive — a community contest with a publicly available rubric and recognition incentives — could produce a measurable improvement in the quality of LLM-assisted physics submissions relative to a pre-established baseline. Across all six rubric categories and the normalized composite score, H_1 was supported. The contest arm produced a mean S_{norm} of 48.9 against a baseline of 28.2, with a noise-corrected Hedges' $g' = 1.33$ (95% CI: [0.60, 2.33]) on the composite and large effects in five of six individual categories. These results were obtained under a pre-registered methodology that explicitly modeled within-model scoring variance rather than assuming deterministic outputs at temperature 0. These results were additionally cross-checked against an uncorrected sensitivity analysis; although the initial variance-subtraction approach used here can in principle inflate effect sizes, the observed shifts were negligible in this dataset and did not change any outcome classifications. This nonetheless motivates future work to better characterize scoring noise and refine effect-size calibration.

The finding suggests that structured community incentives represent a viable direction for improving the quality of LLM-assisted amateur science — not by gatekeeping the communities that produce it, but by providing participants with explicit criteria and structured motivation to meet them. The present results warrant further investigation into the scalability and durability of this approach.

AI Usage Disclosure

This study made extensive use of AI tools throughout its development. Models used include GPT-5.2, GPT-5.4 (OpenAI), Claude Sonnet 4.6, Claude Opus 4.6 (Anthropic), and OpenAI's Prism writing tool. These were used to assist with manuscript preparation, code development, and editorial refinement, in addition to the automated scoring described in Section 2.4. The author takes full responsibility for the methodology, analysis, and conclusions presented in this paper.

References

AllHailSeizure. LLMPhysics journal ambitions contest: OPEN. Reddit post, r/LLMPhysics, March 2026. URL `https://old.reddit.com/r/LLMPhysics/comments/1rhod1s/llmphysics_journal_ambitions_contest_open/`.

Buse Atil, Sinem Aykent, Ashley Chittams, Lin Fu, Rebecca J. Passonneau, Elise Radcliffe, Girish R. Rajagopal, Andrew Sloan, Tomasz Tudrej, Ferhan Ture, Zheng Wu, Ling Xu, and Brian Baldwin. Nondeterminism of "deterministic" LLM system settings in hosted environments. In *Proceedings of the 5th Workshop on Evaluation and Comparison of NLP Systems*, pages 135–148. Association for Computational Linguistics, 2025.

John C. Baez. The crackpot index. University of California, Riverside, 1998. URL `https://math.ucr.edu/home/baez/crackpot.html`.

Rodrigo M. Batista and Thomas L. Griffiths. A rational analysis of the effects of sycophantic AI. arXiv, 2026. URL `https://arxiv.org/abs/2602.14270`.

Jeffrey Brainard. arXiv, the pioneering preprint server, declares independence from Cornell. *Science*, March 2026. doi: 10.1126/science.zkb639g.

Mingzhe Cheng, Christine Lee, Pranav Khadpe, Sherry Yu, Dongyeop Han, and Dan Jurafsky. Sycophantic AI decreases prosocial intentions and promotes dependence. arXiv, 2025. URL https://arxiv.org/abs/2510.01395.

Alexis Collier. Vibe physics. YouTube, July 2025. URL https://youtube.com/watch?v=TMoz3gSXBcY.

Jasper Dekoninck, Thomas Gehrunger, Kristinn Rognvaldsson, Chengzhi Sun, and Martin Vechev. BrokenArXiv: How often do LLMs claim to prove false theorems? MathArena, March 2026. URL https://matharena.ai/brokenarxiv/.

Mark Harris. Crackpots and consequences. *Physics World*, 2012. URL https://physicsworld.com/a/crackpots-and-consequences/.

Larry V. Hedges. Distribution theory for Glass's estimator of effect size and related estimators. *Journal of Educational Statistics*, 6(2):107–128, 1981.

Sabine Hossenfelder. What I learned as a hired consultant for autodidact physicists. Aeon, 2016. URL https://aeon.co/ideas/what-i-learned-as-a-hired-consultant-for-autodidact-physicists.

Steve Rathje, Mengyu Ye, Laura K. Globig, Riya M. Pillai, Vitor O. de Mello, and Jay J. Van Bavel. Sycophantic AI increases attitude extremity and overconfidence. PsyArXiv, September 2025. URL https://doi.org/10.31234/osf.io/vmyek_v1.

r/LLMPhysics. r/LLMPhysics. Online community, Reddit, 2025. URL https://www.reddit.com/r/LLMPhysics/.

r/LLMPhysics. r/LLMPhysics subreddit snapshot. Internet Archive, March 2026. URL https://web.archive.org/web/20260324190754/https://www.reddit.com/r/LLMPhysics/. Archived on March 24, 2026.

Mrinank Sharma, Meg Tong, Tomasz Korbak, David Duvenaud, Amanda Askell, Samuel R. Bowman, Newton Cheng, Esin Durmus, Zac Hatfield-Dodds, Scott R. Johnston, Shauna Kravec, Timothy Maxwell, Sam McCandlish, Kamal Ndousse, Oliver Rausch, Nicholas Schiefer, Da Yan, Miranda Zhang, and Ethan Perez. Towards understanding sycophancy in language models. In *Proceedings of the International Conference on Learning Representations (ICLR 2024)*, 2024.

Ethan Siegel. Why "vibe physics" is the ultimate example of AI slop. Big Think, 2025. URL https://bigthink.com/starts-with-a-bang/vibe-physics-ai-slop/.

Appendix A: Scoring Prompt (guidelines_v1.1)

The following is the complete scoring prompt used for all automated scoring runs, reproduced verbatim from the file `guidelines_v1.1.txt` as stored on disk prior to any scoring. The SHA-256 hash of this file was recorded in the project repository before scoring began, providing independent verification that the prompt was not modified after the study commenced.

```
LLMPhysics Journal Ambitions Contest
Standardized Judge Prompt | v1.1

INSTRUCTIONS

You are a standardized judge for the LLMPhysics Journal Ambitions
Contest. Your sole function is to evaluate the submitted manuscript
strictly against the rubric below. You will not evaluate anything
outside of the rubric criteria. You will not comment on the topic,
the author, the tools used to produce the work, or any other factor
not explicitly listed in the rubric.


RUBRIC

Score the manuscript on the following six criteria. For each criterion,
provide a numeric score within the allowed range and one to two sentences
of justification that directly reference specific content from the
manuscript (quote or paraphrase the paper where possible).

Do not award partial credit outside the allowed range. Do not add
commentary beyond the justification.

---

hypothesis (max 15.0): Evaluates whether the manuscript presents a
scientifically structured hypothesis.
  Full marks: The manuscript states a clear, unambiguous hypothesis AND
  explicitly defines a null hypothesis and an alternate hypothesis. The
  author discusses what specific outcomes would falsify or support each,
  and these failure conditions are mutually exclusive and plausibly
  testable (not requiring physically impossible or science-fiction
  technology). Full marks should only be awarded if a reader could
  plausibly imagine a path toward testing this | the standard is roughly
  "could this pass a grant committee in spirit, even if the tools don't
```

exist yet."
Partial credit: The hypothesis is clearly stated but lacks explicit null/alternate framing, or failure conditions are vague, one-sided, or only discuss confirming evidence.
Low marks: The manuscript has no clear hypothesis, or only asserts claims without any consideration of what would disprove them.

novelty (max 15.0): Evaluates whether the manuscript makes a genuine, demonstrable intellectual contribution.
Full marks: Author cites recent relevant work (past ten years), explicitly identifies gaps or limitations in those prior approaches, and clearly articulates how their proposal differs from and improves upon them. It is the author's responsibility to make this case | the judge does not perform an independent literature review. Claiming novelty without doing this comparative work does not earn full marks, even if the idea happens to be genuinely novel.
Partial credit: Author attempts to position their work against existing literature but citations are sparse, outdated, or the connection between prior work and their approach is unclear or underdeveloped.
Low marks: Author claims novelty with little or no citation support, or makes no meaningful attempt to engage with existing work in the field.

scientific_humility (max 15.0): Evaluates whether the author demonstrates appropriate restraint, proper attribution, and honest acknowledgment of the boundaries of their work.
Full marks: Author clearly delineates the scope and limitations of their proposal. Claims are proportionate to the evidence presented | they do not exceed what the paper actually supports. Author discusses what their work does and does not address, without speculative overreach, boasting, or challenging tone. Does not dismiss established theory as fundamentally flawed without substantive evidence. Properly cites and attributes prior work and foundational concepts | does not present existing ideas as their own. Explicitly identifies open questions and areas requiring further investigation.
Partial credit: Author acknowledges some limitations and may identify some open questions, but claims occasionally exceed the scope of evidence presented, or the boundaries of applicability are unclear. May include minor instances of boastful language or unfounded criticism of established theory. Citation of prior work is incomplete or inconsistent.
Low marks: Author makes grandiose claims unsupported by the work shown, fails to acknowledge significant limitations or open questions, adopts a boastful or challenging tone, dismisses well-established theory as fundamentally wrong without substantive evidence, or fails to properly attribute prior work and foundational concepts.

engagement (max 20.0): Evaluates whether the author demonstrates genuine, substantive engagement with the current state of their field | not just citation presence, but meaningful connection between cited work and the proposal.

Full marks: Author demonstrates substantive engagement with recent literature in their specific subfield. Citations are relevant and directly connected to the work presented. Author explicitly articulates how their proposal relates to, extends, or challenges current work. Does not rely on foundational classics without also engaging with modern developments in the same area | if a foundational paper is cited, recent work in that area should also be present. It is the author's responsibility to make these connections clear | the judge does not infer unstated relevance.

Partial credit: Author cites relevant work but engagement is somewhat superficial | citations may be present but lack clear connection to the proposal, or the author relies too heavily on old foundational work without recent context. Connections between cited work and the proposal are implied rather than explicitly stated.

Low marks: Author uses citations as decoration without substantive connection to their work, cites papers that share keywords but are not genuinely relevant to the specific problem being addressed, or demonstrates minimal engagement with the current state of the field.

rigor (max 10.0): Evaluates whether the author's mathematical and logical work is both valid and sound | not just formally correct, but physically justified.

Full marks: Author presents mathematically sound derivations with correct dimensional analysis and proper notation. Premises are physically justified and grounded in established physics or clearly stated assumptions | not invented concepts. Logical flow from premises to conclusions is explicit and rigorous. Claims follow necessarily from the work shown. Quantitative claims about physical properties or behavior are supported by mathematical derivation or calculation. It is the author's responsibility to show this work | the judge does not infer or reconstruct missing derivations.

Partial credit: Author attempts rigorous derivation but has minor mathematical or dimensional errors, or premises lack full justification. Logic is mostly sound but may have gaps or unstated assumptions. Some claims may lack full mathematical support.

Low marks: Author presents equations or logic without proper derivation, uses unjustified or physically implausible premises, has significant mathematical or dimensional errors, logical flow is unclear or broken, or makes quantitative claims about physics without mathematical backing.

citations (max 10.0): Evaluates whether citations are recent, relevant,

```
properly attributed, and genuinely connected to the work | not padding
or decoration.
  Full marks: Citations are recent (preferably within past ten years),
  properly formatted consistently throughout the manuscript, and directly
  relevant to the specific claims being made. Author explicitly connects
  citations to their work in the body of the text | not just listed in a
  bibliography. Citations are woven throughout the manuscript, supporting
  claims as they are made. If older foundational sources are cited, the
  author provides clear justification for why recent work is insufficient.
  The more thoroughly citations are used and connected to the work, the
  better. The judge does not infer connections between citations and claims
  | if the author does not explicitly make the connection, it does not
  exist.
  Partial credit: Citations are mostly recent and relevant but may lack
  clear connection to some claims, or formatting is inconsistent. Some
  older sources may be present without clear justification. Citations are
  present but not consistently woven throughout the work.
  Low marks: Citations are sparse, outdated, or lack clear relevance to
  the work. Author makes claims without citing supporting sources. If any
  citation is fabricated or clearly false, the author receives zero points
  for this entire category regardless of other citations present.

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<output_requirements>
Return ONLY one valid JSON object.
Do not include markdown code fences.
Do not include explanatory text.
Do not include text before or after the JSON object.
Do not include trailing commas.
Scores must be numeric values formatted with exactly one decimal place
(e.g., 12.0, 11.5, 9.0).
Your entire response must be parseable by a standard JSON parser.
</output_requirements>

<json_schema>
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  "novelty": { "score": 0.0, "justification": "" },
  "scientific_humility": { "score": 0.0, "justification": "" },
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}
</json_schema>
```

SUBMISSION

The manuscript to be evaluated is provided as an attached document.